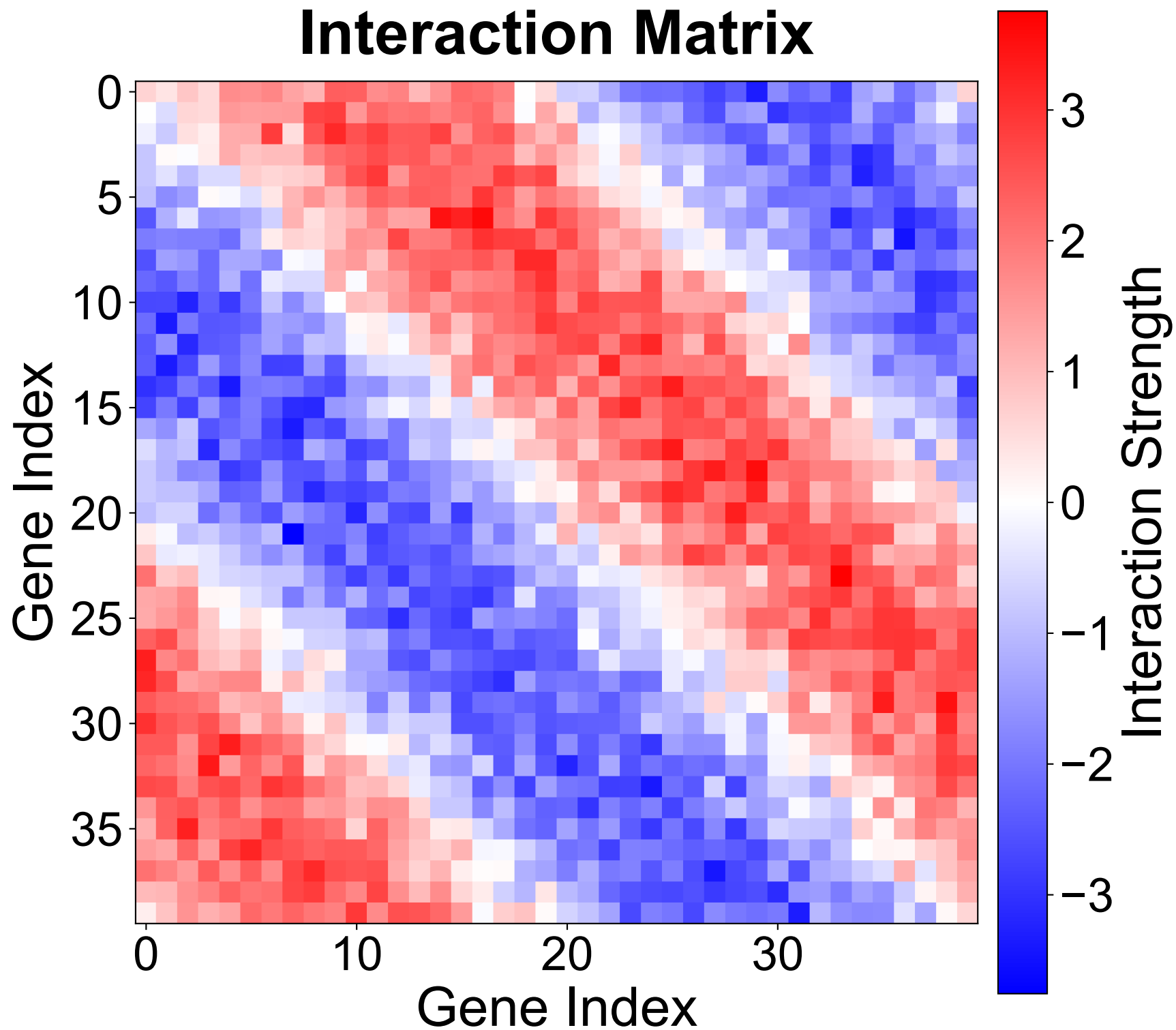


# Non-linear transcriptional activation model:

$$\frac{d\mathbf{r}}{dt} = \sigma(\mathbf{A}\mathbf{p}; \alpha, \tau) - \mathbf{B}\mathbf{r}$$

$$\frac{d\mathbf{p}}{dt} = \mathbf{C}\mathbf{r} - \mathbf{D}\mathbf{p}$$

**Ground Truth  
Interaction Matrix**



**Estimated  
Lead-Lag Matrix**

